

Aryaroop Majumder

Machine Learning Engineer | Model Development, Evaluation and Data Pipelines | Python, PySpark, AWS

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TARGETED SUMMARY

Machine Learning Engineer with 3+ years of applied AI/ML, analytics, model evaluation, **PySpark** data engineering, cloud deployment, documentation, and stakeholder-facing delivery experience. Seeking to apply **transferable ML engineering strengths** to **credit model development** work involving **model lifecycle** support, validation evidence, **implementation testing**, monitoring, documentation, and **root-cause analysis**. Hands-on with **Python, PySpark, SQL, AWS Glue, AWS Lambda, Airflow, Docker**, CI/CD, statistical analysis, classification models, **model performance** monitoring, and communicating technical findings to business and engineering stakeholders.

TRANSFERABLE FIT FOR CREDIT MODEL DEVELOPMENT

Modeling and Statistics: Hands-on supervised **machine learning**, classification models, feature engineering, segmentation, error analysis, **root-cause analysis**, explainability, and basic/advanced ML algorithms; familiar with **logistic regression**, probability scoring, calibration, scorecard-style modeling concepts, and champion/challenger evaluation patterns.

Model Evaluation and Validation: Hands-on **model-quality review** and **performance analysis**; familiar with **AUC, ROC, KS, Gini**, confusion matrix, **precision/recall, lift/decile analysis, PSI**/stability monitoring concepts, **drift analysis**, back-testing concepts, **implementation testing**, validation evidence, annual review support, and **model performance** KPI documentation.

Credit and Risk Context: Foundational familiarity with consumer credit lifecycle concepts including acquisition, account management, collections, credit bureau data, credit policy, model governance, regulatory review support, risk controls, audit-ready documentation, and clear written/oral communication.

Tools and Platforms: **Python, PySpark, SQL**, Pandas, NumPy, scikit-learn-style ML workflows, **AWS Glue, AWS Lambda, Amazon EC2, Amazon S3, AWS** data pipelines, SageMaker familiarity, SAS basic familiarity, **Airflow, Docker, GitHub Actions**, Azure, GCP, **data warehouse** and **analytic data systems**.

PROFESSIONAL EXPERIENCE

Data Science Specialist | Brainlabs | Bengaluru, India | April 2025 - Present

- Designed and delivered **SearchIQ** analytics workflows spanning feature extraction, classification logic, model evaluation, documentation, stakeholder demos, and production-style delivery controls, reducing SEO analysis turnaround from months to weeks and cutting analysis time by **85%**.
- Built and improved **topical classification** and **keyword-analysis** modules using embeddings, NLP techniques, validation workflows, and error analysis to improve keyword analysis accuracy to **95%**, translating ambiguous business problems into measurable analytics outputs.
- Developed a client-agnostic classification module for heterogeneous search phrase sources after solution evaluation, architecture option analysis, technical grooming, and stakeholder review, reducing classification cost by **60%** and turnaround time by **30%**.
- Used **LLM evaluation frameworks** and **model-quality review** patterns to assess groundedness, context relevancy, hallucination risk, answer quality, and retrieval quality, strengthening transferable **model validation**, monitoring, evidence collection, and performance-deterioration analysis skills.
- Built GenAI product-tagging automation for a **Fortune 500** global consumer products client by combining image signals, content scraping, prompt engineering, classification logic, validation checks, and review workflows to reduce manual categorization effort and create a scalable decision workflow.
- Prepared executive decks, business one-pagers, metric narratives, and stakeholder demos for senior leadership and client teams, communicating model behavior, limitations, operational value, and next-step recommendations in a clear, concise manner.
- Led **Docker, GitHub Actions, Airflow, LangGraph**, and AI knowledge-sharing sessions, supported repeatable workflow design, created documentation, reviewed technical approaches with data engineering peers, and helped convert prototypes into governed delivery routines.

Associate ML Engineer | Prescience Decision Solutions | Bengaluru, India | August 2024 - March 2025

- Developed scalable **PySpark** ETL and data transformation workflows for **Amazon** using **Python**, Pandas, and **PySpark**, supporting reusable feature-ready datasets, schema normalization, data preparation, and downstream analytic/modeling workloads.
- Implemented data quality products with rule-based validation, exception handling, audit-friendly reporting, and quality controls, resolving **99.99%** of identified data quality issues across supported workflows and strengthening model-input reliability.
- Engineered big data production pipelines and optimized **AWS Glue** DPU usage, improving processing performance by **35%** and reducing operational costs by **50%** while maintaining stable execution for large-scale data workloads.
- Supported **AWS** and Azure deployments across **AWS Lambda, AWS Glue, Amazon EC2, Amazon S3**, Azure Blob Storage, Azure VM, and Azure OpenAI-enabled environments, partnering with engineering teams on release readiness, debugging, and production stability.
- Created technical documentation, implementation notes, data validation summaries, and stakeholder updates for analytics products, supporting review cycles, delivery transparency, and operational handoff quality.

Data Science Intern | Prescience Decision Solutions | Bengaluru, India | June 2023 - July 2024

- Built a **Flask API**-based data quality product deployed across **AWS** and Azure environments, then optimized validation logic, backend performance, and accuracy by **30%** for production-oriented analytics workflows.
- Developed an **Airflow**-based ML/data pipeline using open-source orchestration for scheduled ingestion, transformation, execution monitoring, and repeatable pipeline delivery, reducing recurring platform costs and increasing processing speed by **25%**.
- Built a Natural Language to **SQL** chatbot with Llama 3.1, Google Flan-T5, StarCoder, **Streamlit, Flask API**, and **Docker**, translating business-language questions into structured data queries and supporting analytic data-system usability.

SELECTED PROJECTS

- **Model Evaluation and Retrieval Analytics - SearchIQ:** Designed evaluation workflows around keyword classification, embedding similarity, topic clustering, retrieval quality, groundedness checks, and stakeholder-ready metric reporting, building transferable skills for **model performance** monitoring and deterioration analysis.
- **Hybrid Contextual RAG Chatbot** - GitHub: Built a production-style RAG application with **FastAPI** ingestion/Q&A endpoints, multi-format document ingestion, **OCR** extraction, **Qdrant vector search**, **BM25**, **dense retrieval**, **reciprocal rank fusion**, **reranking**, grounded answers with citations, **Docker** deployment, and **GitHub Actions CI/CD**.
- **Dynamic Graphing System** - GitHub: Built a Dockerized **Streamlit** analytics dashboard for interactive business analysis, data visualization, and lightweight metric exploration, demonstrating dashboarding and data communication skills relevant to model monitoring.
- **Natural Language to SQL Assistant** - Prescience: Built a conversational database access prototype using open-source LLMs, **Flask API**, **Streamlit**, and **Docker** to help non-SQL users interact with structured data and **analytic data systems**.

EDUCATION

Lovely Professional University | B.Tech, Computer Science Engineering, Data Science, Machine Learning and AI | 2020 - 2024 | Phagwara, India

CERTIFICATIONS AND AWARDS

- **Amazon ML Summer School 2022** - Top 200 across India
- Building Generative AI Solutions with Best Practices and Design Choices - Technical, Level 300 Advanced ML (**AWS**)
- Introduction to Artificial Intelligence (AI) (**IBM**)
- Generative AI: Introduction and Applications (**IBM**)
- Introduction to Model Context Protocol (**Anthropic**)
- Certificate of Completion: Claude 101 (**Anthropic**)